SCSE2233 Software Design & Architecture — Final Exam

Q1(a) Model Answer: WBS + Linear Responsibility Chart (6 marks)

Case study: University Student Portal | 5 team members (S1–S5)

What the question is really asking (plain words)

Cut the whole project into a tree of small tasks — that is the **WBS**. Then make a table showing which of the 5 people does each task — that is the **LRC**. Codes: **1** = main owner, **2** = helper, **3** = checker.

Your WBS must satisfy THREE criteria (write these words to earn marks)

1. Scope accuracy	Put TASKS ONLY — no dates/durations, no arrows, no people's names inside the WBS.
2. Completeness	Cover the whole design job: architectural design + detailed design + documentation (with reviews).
3. Level of detail	Break down until each bottom task is small enough to give to one person and estimate.

STEP 1 — Work Breakdown Structure (write this list)

Outline #	Task Name
1	**Student Portal Design Project**
1.1	**Architectural Design**
1.1.1	Evaluate Alternative Architectures
1.1.1.1	Prioritize Design Objectives (security, performance, availability)
1.1.1.2	Formal Review of Evaluations
1.1.2	Select Architectural Design
1.1.2.1	Make Initial Selection
1.1.2.2	Formal Review to Select Final Design
1.2	**Detailed Design**
1.2.1	Design Academic Records Module
1.2.2	Design Assignments & Course Materials Module
1.2.3	Design Announcements Module
1.2.4	Design Role-Based Dashboards & Secure Access
1.3	**Documentation**
1.3.1	Write Architectural Design Document
1.3.2	Formal Review (Architectural Design)
1.3.3	Finalize Architectural Design Document
1.3.4	Write Detailed Design Document
1.3.5	Formal Review (Detailed Design)
1.3.6	Finalize Detailed Design Document

STEP 2 — Linear Responsibility Chart (same tasks + 5 people)

Outline #	Task Name	S1	S2	S3	S4	S5
1.1.1.1	Prioritize Design Objectives	1	2	3		
1.1.1.2	Formal Review of Evaluations	1		3		3
1.1.2.1	Make Initial Selection	2	1		3	
1.1.2.2	Formal Review of Final Design	3	1			3
1.2.1	Academic Records Module		2	1		3
1.2.2	Assignments & Course Materials			1	2	3
1.2.3	Announcements Module	3		2	1	
1.2.4	Dashboards & Secure Access	2			1	3
1.3.1	Write Architectural Document	3			2	1
1.3.2	Formal Review (Architectural)	1	3			3
1.3.3	Finalize Architectural Document		2		1	3
1.3.4	Write Detailed Document		3	2		1
1.3.5	Formal Review (Detailed)	1		3	3	
1.3.6	Finalize Detailed Document			3	1	2

1 Primary responsibility (the owner — exactly ONE per row)

2 Support role (helps)

3 Review role (checks)

How to earn all 6 marks (self-check before you move on)

✓ Hierarchical numbered outline 1 → 1.1 → 1.1.1 → 1.1.1.1 (2 marks).

- ✓ Tasks only — no durations, no arrows, no names inside the WBS = scope accuracy (1 mark).
- ✓ Whole lifecycle shown: architectural + detailed + documentation with formal reviews = completeness (1 mark).
- ✓ Bottom tasks small enough to estimate = level of detail (1 mark).
- ✓ LRC shares ALL tasks over the 5 people, and every row has exactly ONE '1' (1 mark).

Traps that kill marks (the Doctor checks these)

- Two '1's on the same row = no single owner = you LOSE the last mark. Every row = one 1 only.
- Do NOT draw a Gantt chart or write dates — that is a different tool and breaks scope accuracy.
- Do NOT stop at level 1.1 / 1.2. Go down to the small tasks or you lose 'level of detail'.

Tip: in the exam you may draw the WBS as a tree (boxes) OR as this numbered list — both get full marks. The list is faster to write by hand.